DDCS V4.1 · SEA OTTER

User manual

Replacement controller software for the DDCS V4.1 — daily use at the machine



github.com/JSFrouws/ddcs-4.1-Sea-Otter · docs/user-manual.md

For the person at the machine. Sea Otter is the replacement controller software for the DDCS V4.1 (`app/`). This manual covers daily use. Design background is in operator-procedures.md, the network API in api-reference.md.

> **Not yet tested on a machine with drives connected (2026-10-21).** Tested on a real controller with no drives or motors attached: step-exact positions, real CAM programs up to 64 000 lines, limit inputs and controlled stops through the real input path. Do the first real tests with the motors unpowered or able to move freely, with a hard-wired E-stop that cuts the drives. No warranty, no liability: see the README.

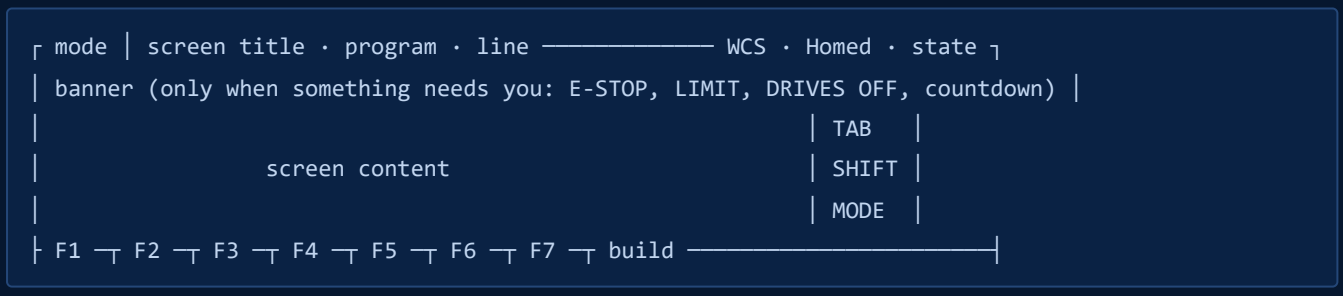
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0. Installing

From a USB stick, no PC tools needed on the controller: usb-install.md. In short: copy the `ddcsv4` install folder to a FAT32 stick, set the IP address in `network.conf`, plug it in, switch on, then take the stick out. The browser panel and the API answer at `http://<that IP>:8080/`. This is the controller's own update route: the vendor's manual (section 3.4.3) updates its software the same way.

Back to the original software: download the newest DDCS V4.1 software from Digital Dream at www.ddcnc.com › Downloads › Software & firmware (newest at the time of writing: *DDCS V4.1 (2025-04-04) software*), unpack it, copy its `ddcsv4` folder to an empty FAT32 stick, power off 10 s, insert, power on, wait ~30 s, remove the stick. Details in usb-install.md. Remote access with a password: ssh-setup.md.

1. The screen



- **Top right**: active work offset (G54…), `Homed ✓/–`, and the machine **state** (READY, RUN, HOLD, TOOL CHANGE, E-STOP, LIMIT, DRIVES OFF, SIMULATE…).
- **Soft keys F1–F7** along the bottom: their labels change per screen. **F7 is always Page R**: it opens the **screen menu** (below). With **Shift** lit the labels turn amber and show the second functions.
- **Right strip**: what **Tab**, **Shift** and **Mode** do right now.
- **Toasts** (white box near the bottom): the result of what you just did, or why a key was refused. Nothing is refused silently.

2. Keys

Key	Does
Mode	Run ↔ Jog. Jog is locked while a program runs or holds (not during a tool change).
F7 Page R	Opens the **screen menu**: the soft keys become the screens, press one to go there (see below). Page R again or Cancel closes it
Shift + F7	Back to the last screen
Tab	Next pane on the screen (e.g. Toolpath ↔ Program)
Shift	ALT layer for the next key (lit while active): touch-off, second functions (below)
Y± / X±	Move in lists and tables; jog in Jog
Enter / Select / Modify	Open / accept / edit the focused item
Cancel	Back / discard
F1 Start	Start or resume the program
F2 Pause	Feed hold on **every** screen while something moves
F3 Reset	Stop on **every** screen. Also switches the drives on after DRIVES OFF

The screen menu (Page R, then):

F1	F2	F3	F4	F5	F6	F7
Run	Macros	Tools	Files	Work	Settings	close

Jog isn't in the menu: **Mode** goes there directly. While something moves, F2 and F3 stay **Pause** and **Reset**, menu or not. The menu closes after any key, or by itself after 8 s.

Second functions (Shift, then the key):

Screen	F1	F2	F3	F4	F5	F6
Run	Start from the line picked in the listing	**WCS**: next work offset (G54 → G55 …)	Rewind to line 1 (asks first)	View: fit / reset	**Goto line** (below)	Coolant on/off
Jog	Home	**Safe Z** (Z up only)	Drives	Step size back	WCS back	Coolant on/off
Work				**Job: none** (clears every JOB mark)		

A running program still pauses with F2 and resets with F3, Shift or not.

3. Daily procedure

3.1 Power up

1. Switch on. If the E-stop is pressed, release it. The banner tells you what's needed.
2. **Mode** → Jog. **F1 Home** if you want soft limits (not required). Reset stops homing.

3. *Optional:* **Keep the homed state after a restart** (*Settings › Homing & limits*, off by default). The position is saved 2 s after every stop. After a restart or power cut, Sea Otter is still homed if the machine was at rest when the power went. It is not homed if the power went while it was moving. For a quick reboot when you don't need the last hundredth: the drives were reset too (a small *clunk* is normal), so expect up to 1/100–1/50 turn, **0.05–0.1 mm on a 5 mm screw**. Home again for precision work.

3.2 Load a program

1. **Page R** → Files. **F3** switches Local / USB. **Y±** picks a file.
2. **F1 Load** (or Select). The whole file is checked first; an unsupported code is reported with its line number and the file is not loaded. Supported: G0–G3 (also in the G17/G18/G19 planes), G4, G10 L2/L20, G20/G21, G28, G53, G54–G59, drilling cycles G73/G81/G82/G83/G85/G89 with G80 and G98/G99, G90/G91, G94, M0/M1, M3/M4/M5, M6 (pause for a tool change), M7/M8/M9, M10/M11, M30. G40/G49/G61/G64 are accepted. G43 is accepted but ignored (no tool length compensation). G41/G42 (cutter compensation) and G84 (tapping) are refused.

3.3 Zero the workpiece

1. Jog to the part's edge (Jog: arrow keys; **F4 Step** picks 0.01 / 0.1 / 1 / 10 mm or continuous).
2. **Shift + the axis key towards the part** sets that axis: with an edge Ø set (Jog setup, or Tools › F3) it accounts for the radius (X+ edge → X = −Ø/2). Press the same key again to set 0 instead.
3. Z: touch the top, **Shift + Z−** → Z = 0.
4. Check with **Goto 0** (Jog F2): Z up to safe height, then XY to 0. A blue banner shows it while it moves. **F2 Pause** holds it and **F1 Start** continues it, from any screen. **F3 Reset** ends it. **Shift + F2** on Jog is *Safe Z*: Z up only.
5. Several parts: **Jog F5 WCS** picks G55, G56…; on **Work** mark them **F4 In job**. Start then cuts **per tool**: everything the first tool does, on every JOB part in turn, then **one** tool change, then the second tool on every part, and so on. So 4 parts with 3 tools means 3 tool changes, not 12. Between parts Z goes up to the safe height. **Shift + F4** on Work clears the job.

3.4 Check, then run

F5 TryCut on Run cycles what Start will do:

Armed	What happens
Run	The real job
TryCut	Spindle off; Z goes once to the **clearance level** (work Z0 + Safe Z) and stays there. The machine flies the whole path in the air
Simulate	Nothing moves. The position readout turns **orange**: those coordinates are fictitious. Afterwards the real position is shown again. M0/M1/M6 pauses still stop it, just like a real run

F1 Start. A program started over the network counts down on the panel first (default 5 s); **Reset cancels**.

3.5 During the run

- **Overrides**: pendant switch OFF + hold the dead-man + turn the wheel: **X1 = feed**, **X10 = spindle** (2 % per click), **X100 = feed in 10 % steps**. Feed and spindle go up to **300 %**. The bars stop at 200 % and the number shows the rest in amber. **Rapid follows the feed down** and never goes above 100 %. Its bar is half width, so 100 % lines up on all three bars.
- **The speed is capped** at what the controller can actually deliver (960 mm/min per axis at 2000 steps/mm). A higher F or override runs at the cap.

- **F4 View** cycles **2D → ISO → CPU saver**. CPU saver hides the toolpath and redraws once a second, leaving more time for motion. After an E-stop or limit the view switches to ISO by itself.
- **The preview** always fits and centres the whole picture: the program, the work zero's axes, and small axes labelled G55, G56… at the zero of every other JOB part. The thin box around the toolpath is the program's size (there is none without a program). The grid lies on the **machine's Z floor** (the Z soft limit −) in the ISO view. Its coarse lines are 1 cm when the picture is under 10 cm, 5 cm up to 50 cm and 10 cm up to 1 m. Its fine lines, barely lighter than the background, are a tenth or a fifth of that. Both steps are shown bottom right.
- **No program loaded**: the listing says so. Enter or Select picks nothing.
- **M0 / M1**: the machine holds with a message; **F1 Start** continues.
- **M6** (tool change): see § 3.7.
- **F2 Pause** = feed hold (F1 resumes). **F3 Reset** stops; the lines done so far are remembered.

3.6 Resume after a stop

Start continues where the program stopped. After Reset, an E-stop or a limit, the stopped line becomes the resume point. Start rebuilds the state there (WCS, F, S, tool, coolant) and approaches it: Z up, XY across, then down.

- **Back to the beginning** takes **two** Resets. The first asks *"Go back to the BEGINNING (line 1)?"* in an amber banner. A second Reset within 5 s confirms; any other key keeps the line.
- **Another line**: Run → **Tab** → Program, pick it (**Y±** one line, **X±** ten), **Select**. Or **Shift + F5 Goto line**: type the line number in the digit editor. Beside it, a list shows good places to continue from. These are the program's **sections** (comment lines that start a block, like CAM operation names). Without sections, it lists the **retracts to its usual safe height** (the highest Z it goes up to most often), each with its line number. Enter goes there, Cancel doesn't.
- **After a restart or a power cut** the program comes back loaded, with the resume point near where it was. While a program runs, Sea Otter saves its line once a minute, so the resume point can be up to a minute behind. Check the line before you press Start. It never starts by itself.

3.7 Tool change (M6)

Settings › Tool change › On M6 chooses what an `M6 T…` in the program does:

Pause: change by hand (default): 1. The spindle stops, Z goes up to the **safe height** (or, with *Change the tool at › Tool-change position*, to that position in machine coordinates; needs a homed machine). 2. The state becomes **TOOL CHANGE** and the amber banner says which tool to put in. 3. **Jogging is allowed**: **Mode** → Jog, jog, change the tool, and touch off Z with the new tool (**Shift + Z−**) if its length differs. In a multi-part job, a Z touch-off on one part is applied to the other parts too: it's the same tool. 4. **F1 Start** shows **"Back to the program?"**: a straight line to the position the program stopped at (at least the safe height, then straight down if lower), the speed, and the tool. **F1 Start or Enter** goes, **Cancel** stays. Jogging after the question cancels it: Start asks again with the new route. 5. The return speed is *Settings › Tool change › Return at* (% of rapid, default 50 %). The spindle and coolant come back on as the program had them before the return move starts.

F3 Reset during a tool change stops the program. Its progress is remembered, as with any stop. If the tool asked for is already in, nothing happens. Simulate never stops for a tool.

Run a tool-change macro: Sea Otter runs a G-code file instead of stopping, then goes back to the program (Z up first, then across, then down). It looks in the macro folders (`/local/macros` first, then `/ddcsv4`) for: - `toolchange-T3.nc`: **one file per tool** (copy one and adjust it per tool), or else - `toolchange.nc`: **one file for all tools**, using `{tool}` for the tool number.

Besides the macro placeholders (§ 5), a tool-change macro can use `{tool}` (the new tool), `{tool.prev}` (the one in now), `{tool.dia}` and `{tool.len}` (from the Tools table). Example:

```
(name: Tool change)
(info: to the rack, slot = tool number)
(needs: homed)
G90
G53 G0 Z{macro.tc_z}
G53 G0 X{macro.tc_x} Y{macro.tc_y}
M0 (put in T{tool}, then Start)
```

Without either file, it falls back to changing by hand.

Ignore M6: the tool number is noted and the program runs on.

4. When the machine stops on its own

4.1 Controlled stop (no lost steps)

Settings › Homing & limits › Controlled stop on Reset, E-stop, limit — **on by default**. Reset, the E-stop and a limit switch all **brake along the path** at the axes' maximum deceleration, so the motors don't lose steps and the position stays valid: no re-homing, no re-zeroing. The machine runs on by the **braking distance**: about **1 mm** at full speed (measured). **Mount limit switches with at least that much room before the hard stop.** Switch it off and the stop is abrupt (≈ 0.3–0.5 mm), but a fast-moving motor may then lose steps.

4.2 E-stop

The E-stop is for **unexpected or unwanted movement**. It stops motion, spindle and outputs. **The drives stay on**, so the axes keep their position and holding torque. Release the button and the machine is ready again. Jog, or resume from a line. **To remove power, switch the machine off.**

4.3 Limit switch

The banner shows e.g. `X+ LIMIT`. Only that direction is blocked. **Mode** → Jog and:

- **back off** the switch (X− here): the limit clears by itself when the switch releases;
- or move **sideways** (Y) or **up/down** (Z): never blocked.

Two or three switches at once (e.g. driving into a corner): each one stops the move and blocks its own direction; the banner lists them all (`X+ Y− LIMIT · jog X− Y+ to release`). LIMIT only goes away when the **last** switch is released.

A program that was running has stopped with its progress remembered: resume as in 3.6. Which input each axis's switch is on: *Settings › Homing & limits › X/Y/Z limit/home switch input* (this machine: IN02 / IN03 / IN04; 0 = no switch on that axis).

4.3a Soft limits and bench limits

Two software boxes keep the machine inside its travel, without touching a switch:

	In force	Set in *Settings › Homing & limits*
Soft limits	once the machine is **homed** (before that, positions mean nothing). Default: the machine's **400 × 200 × 100 mm** (X 0…400, Y 0…200, Z −100…0 below the Z switch)	*X/Y/Z soft limit − / +*, switch: *Soft limits*
Bench limits	**always**, homed or not: a work box, e.g. the part of the table you use, or a safe box on the bench while testing	*X/Y/Z bench limit − / +*, switch: *Bench limits*

When both are on, the tighter one counts. What they do:

- **A program that would leave the box is refused** at Start — rapid moves count, not only cutting moves (a `G28` to machine zero can therefore be refused) — with the axis, the side and how far: e.g. *Program goes past the X+ bench limit on G54: reaches 675.8 mm, limit 587.8.* TryCut is checked at its clearance height. Simulate still runs (it moves nothing) and says the program is outside.
- **A jog towards the edge slows down and stops exactly on it.** A step jog past the edge is shortened to end on the edge; a held (continuous) jog brakes at the axis' acceleration as it gets close. Measured on the controller: from 16 mm/s to a standstill within the last ~0.7 mm, landing on the edge to the step. The next jog that way is refused (*X+ is at its bench limit*); the other way is free.
- Goto 0, Park and Tool change are refused if their target is outside the box.

4.4 DRIVES OFF

Only when someone switched the drives off (Jog F3, or over the network). The banner says who. **F3 Reset** on any screen switches them on again.

5. Macros

Macros are **small G-code files** — nothing is built in. The Macros page lists every macro file it finds, and you can add, change or remove them yourself.

Running one: Page R → Macros, **Y±** pick, **Select / Enter / F1**. The macro is *armed*: Sea Otter switches to the **Run page**, the program pane shows the macro's lines and a teal banner says *MACRO · Park armed*. Then it behaves exactly like a program:

Key	Does
F1 Start	runs the macro (its values are filled in at this moment, see below)
F2 Pause	holds it; **F1 Start** continues
F3 Reset	stops it halfway (or cancels it before Start)

When the macro ends — finished, Reset, E-stop or a limit — **your loaded program is back**, with its resume line and its Run / TryCut / Simulate setting as they were. A macro that would leave the soft or bench limits is refused at Start, like a program.

Where macros come from:

Folder	What
`macro-*.nc` in the app folder (`/ddcsv4`)	the package's set; installed and updated with the USB package. The package brings: Safe Z, Go to work zero, Park, Tool change position, Spindle warm-up, Zero all
`macros` folder on the **CNCDISK** network share (`/local/macros`)	**your own**: copy `.nc` files there from a PC, no USB stick or restart needed (F4 Rescan on the Macros page)

Writing one: plain G-code, with optional header comments on the first lines, and `{placeholders}` that are filled in when you press Start:

```
(name: Park)
(info: Z up, X Y to the park position, Z to its height)
(needs: homed)
G90
G53 G0 Z{max:machine.z,macro.park_z}
G53 G0 X{macro.park_x} Y{macro.park_y}
G53 G0 Z{macro.park_z}
```

`(name: …)`	the name on the Macros page (default: the file name)
`(info: …)`	one line under the name
`(needs: homed)`	refused until the machine is homed (for machine coordinates)
`{jog.safe_z}` , `{macro.park_x}` , …	the value of any setting (the keys in api-reference.md)
`{machine.x}` `{work.z}` …	the current position, machine or work coordinates (x y z a)
`{max:A,B}` `{min:A,B}`	the larger / smaller of two numbers or names, e.g. `{max:work.z,jog.safe_z}` = "Z up to the safe height, never down"

Settings › Macros holds the values the package's macros use (park and tool-change positions, warm-up speed and time); your own macros can use them too.

6. Settings

Page R → Settings. Left: groups, right: values. Quick press / Modify / Enter starts editing. On/off, lists and small numbers change in place with Y± (or hold + wheel); larger numbers open the digit editor. **Enter accepts, Cancel discards.** For a value that may be negative, **Modify** flips the sign (or X− past the leading zero onto the ± cell, then Y±). While the editor is open, the keys that do nothing are greyed out. F2/F3 still pause/reset anything that moves (the edit is dropped). One step past the last value (or before the first) shows an orange ← **back to the groups** line, and the group on the left gets an orange arrow: Enter or a quick press there goes back to the groups, one step the other way returns to the list. Scrolling to the end never leaves the group by itself, and the pendant alone can do everything.

- **Grey values don't apply** right now, and the reason is shown (e.g. "API is off"). They can't be edited until the setting they depend on is changed.
- Values marked **idle only** (machine settings) can't change while anything moves.
- **F2** restores this machine's original values. **F4** exports the settings to USB.

Useful ones: *Display › Default toolpath view* (2D / ISO / CPU saver) · *Jog & touch-off › Safe Z* · *Homing & limits › Controlled stop* · *API* (below) · *Pendant › Reverse wheel direction* · *Tool change* (§ 3.7: by hand / macro / ignore, where, return speed).

API (*Settings › API*): the first line, **ENABLE the REST API endpoint**, switches the whole network API on or off. Under it, each kind of action can be allowed or not. Reading, programs, offsets, tools, settings, overrides and panel messages are on. **Starting programs, jogging, moves (home, Goto 0, Safe Z), spindle, coolant, drives on and panel keys are off** until you switch them on here. These switches can only be changed on the panel. Details: api-reference.md.

I/O overview (Settings, **F6**): every input IN01–IN18 with its level, what is wired to it and whether that function is active (amber). Also shown: the outputs, the spindle voltage, the pendant (E-stop, dead-man, axis switch, multiplier, wheel counts, raw register), the panel keys held right now, the LEDs and the FPGA position counters. **F6** or **Cancel** goes back to Settings.

7. The pendant (optional)

Input	Does
Wheel, dead-man released	**Moves the focus**: list items, settings, digit cursor, view
Wheel, dead-man held, switch OFF	**Changes** the focused value; while running: overrides
Wheel, dead-man held, switch X/Y/Z/4	**Jogs** that axis 0.001 / 0.01 / 0.1 mm per count (refused while running)
Quick press of the dead-man (switch OFF)	**Enter** on the focused item
Jog screen: dead-man held, switch X/Y/Z/4	That axis's row lights up in a soft amber (on Jog and Run) and the jog pane is focused: only then does the wheel jog
Jog screen: dead-man held, switch OFF	The **Jog setup** pane is focused: the wheel changes the setup (step, WCS, …), never jogs
Pendant E-stop	Same as the panel E-stop (§ 4.2)

The wheel never acts as a jog key. Before this fix, turning it on the Jog screen with the switch on OFF and the step on *continuous* started a Y jog that kept going. Also: any key press ends a continuous jog, and touch-off waits until the axis stands still.

Pendant dead, E-stop "pressed"? If every pendant line reads high (I/O screen: raw `0x3FF`), the pendant is unplugged, its cable is loose or it has no power. The E-stop line then reads "pressed" too, and the banner says so. Nothing in the software can bring the buttons back, because they all read "not pressed". The motion controller itself also refuses to step while that line reads "pressed" (measured 2026-10-24), whatever *Pendant › Pendant E-stop enabled* says. Check the plug and the cable.

8. Network

The controller serves a browser panel and the API on port 8080 (`http://<controller-ip>:8080/`). Every panel action is also an API call; see api-reference.md. What the network may do is set in *Settings › API* (§ 6). A remote start, once allowed, counts down on the panel first, and Reset cancels it.

9. Known limits (2026-10-21)

- **Not yet tested with drives or motors connected**, nor with real limit/home switches or a probe (probing is not configured). Limit handling was tested through the real input path with the inputs' level inverted, not with a switch.
- Loading a very large file (64 000 lines) takes ~4 s, and the panel freezes for that time.
- G84 tapping and tool length compensation (G43) are not supported.
- `G28` without axis words sends all axes to machine zero **together**, not Z first, as written in the program. Check CAM output before the first real run.